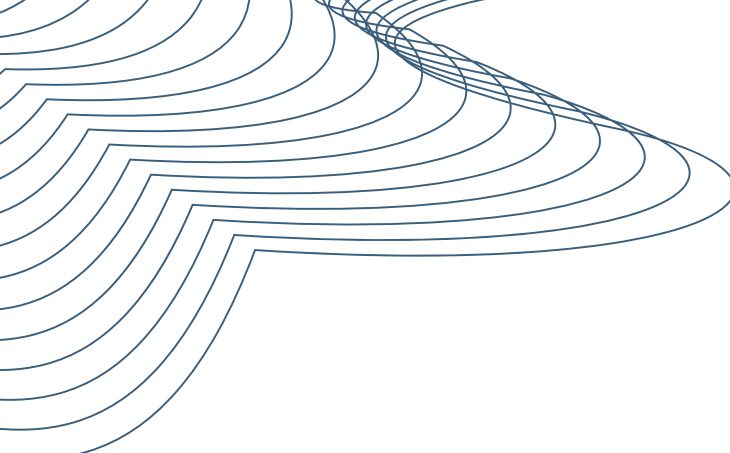




PRIORITY SCHEDULING

OSSP Project

UNDER THE GUIDENCE
OF
SOUMYA



TEAM MEMBERS

2520030204

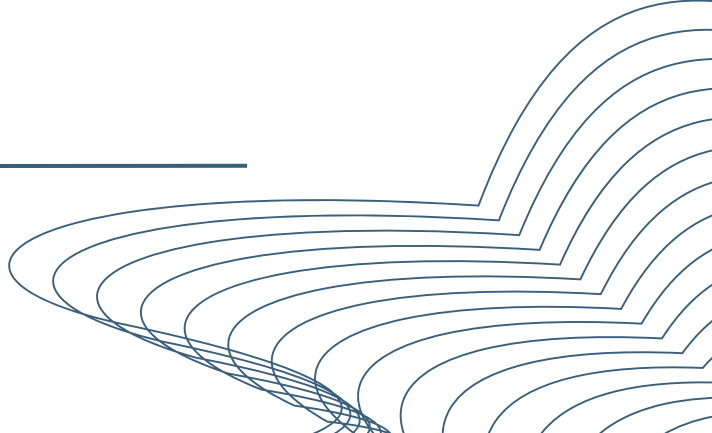
K.Joshi Rushitha

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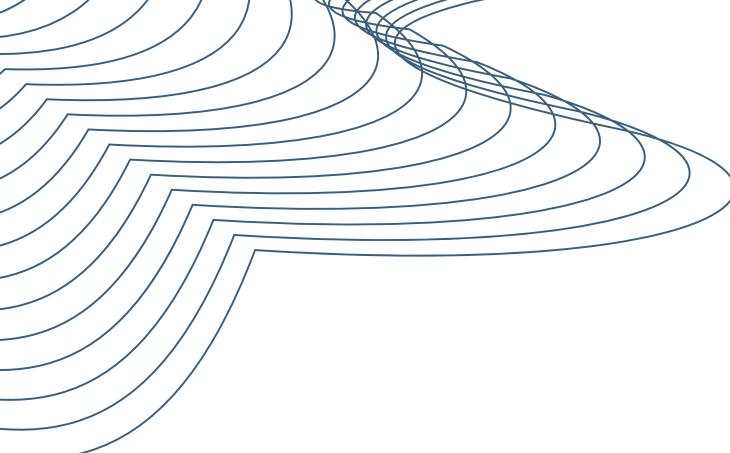




ABSTRACT

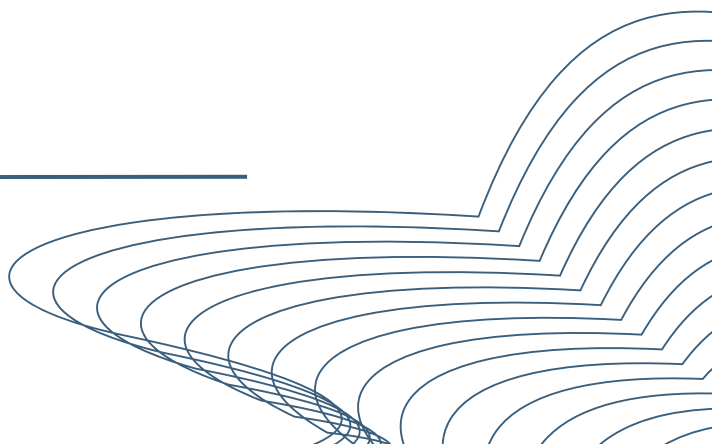


The KL Hospital Priority Management System is designed to manage and organize patients based on their medical priority using Priority Scheduling. The system helps prioritize patients so that those requiring urgent attention can be handled first. It includes patient registration, priority assignment, doctor details, doctor availability, and room allocation. The system also helps identify available doctors and rooms, making the treatment and admission process more organized. Overall, the project aims to improve patient management, resource allocation, and efficient hospital operations.



INTRODUCTION

Priority-based management is important in hospitals to ensure that patients are attended to according to the urgency of their condition. Priority Scheduling assigns a priority level to each patient and helps determine the order of treatment. This project applies Priority Scheduling to hospital management, along with doctor details, doctor availability, and room allocation. It helps organize patient care and manage hospital resources efficiently.



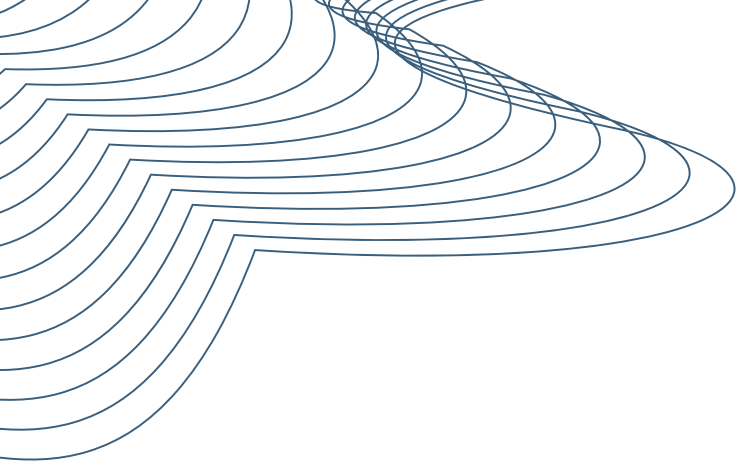
EXISTING SYSTEM VS PROPOSED SYSTEM

Existing System	Proposed System
Proposed System Patient management is mainly manual.	Patient management is priority-based.
Patients may be handled in a general order.	Higher-priority patients are attended first
Doctor availability may be difficult to track.	Doctor availability can be checked easily.
Room allocation is handled separately.	Available rooms can be allocated to patients.
Patient and hospital resources are not centrally managed	Patient, doctor, and room information are managed together.

WORKFLOW / STRUCTURE

The system first records patient details and assigns a priority level based on the urgency of the patient. It then identifies the patient with the highest priority for treatment. The system also checks doctor availability and allocates an available doctor and room when required. This process continues as patients are managed, helping organize treatment and hospital resources efficiently.

Process
INPUT
CHECK PRIORITY
SELECT PROCESS
EXECUTE
CALCULATE WT & TAT
DISPLAY RESULT



SOFTWARE REQUIREMENTS

Software	Purpose
HTML & CSS	To design the hospital website
JavaScript	To add interactive features
Python / Flask	To implement the backend
Linux Terminal	To store patient, doctor and room details



OUTPUT SCREENSHOTS



Patient Information
Please enter accurate patient information.

01

Basic Details
Patient identification information

Patient ID *

Example: P001

Unique Identification number

Patient Name *

Enter full name

Age

Enter age

Gender

Select gender

Contact Number

Enter contact number

Arrival Time *

--:--

02

Medical Details
Information required for priority scheduling

Medical Condition

Describe the patient's medical condition...

Priority Level *

Select according to the urgency of the patient's condition.

1 Low
Nonurgent

2 Medium
Needs attention

3 High
Urgent

4 Critical
Emergency

04

Treatment & Resource Requirements
Used for select doctor and room allocation

Required Specialization

Auto-select from condition

Used for smart doctor matching

Treatment Type

Consultation

☐ Patient requires hospital room / resource

Surgery → Operation Theatre, ICU monitoring → ICU, monitoring → Monitoring Room.

Cancel

Room Management

Track operation theatres, ICU, monitoring and admission rooms.

10

Total Rooms

10

Available

0

Occupied

ROOM	TYPE	STATUS	PATIENT	ACTION
OBS-01	Emergency Observation	Available	—	Maintenance
GEN-01	General Room	Available	—	Maintenance
GEN-02	General Room	Available	—	Maintenance
ICU-01	ICU	Available	—	Maintenance
ICU-02	ICU	Available	—	Maintenance
MON-01	Monitoring Room	Available	—	Maintenance
MON-02	Monitoring Room	Available	—	Maintenance
OT-01	Operation Theatre	Available	—	Maintenance
OT-02	Operation Theatre	Available	—	Maintenance
PRI-01	Private Room	Available	—	Maintenance

KL HOSPITAL

Priority Scheduling

Dashboard

Patient Form

Patient Records

Priority Queue

Doctors

Rooms

Allocation

Scheduling Process

Results

Logout

Administrator

Priority Queue

Higher priority comes first. Arrival time breaks ties only.

Scheduling rule: Priority 4 → 3 → 2 → 1. Within the same priority, earlier arrival time comes first. Use the Emergency button only when the patient has been clinically identified as an emergency.

POSITION	PATIENT	PRIORITY	ARRIVAL	CONDITION	DOCTOR NEED	ROOM	ACTION
#1	p117 keerthi	4 - Critical	01:00	heavy bleeding	General Medicine	Operation Theatre	Emergency
#2	p128 meenakshi	4 - Critical	11:08	heart pain	Cardiology	Operation Theatre	Emergency
#3	p116 ramnadham	4 - Critical	12:00	Accident	General Surgery	Operation Theatre	Emergency

Scheduling Process

Administrator

Priority → Doctor → Room → Treatment.

Core rule: priority is the primary key. Arrival time is used only when two patients have the same priority.

3

Waiting Patients

4 → 1

Priority Order

Doctor

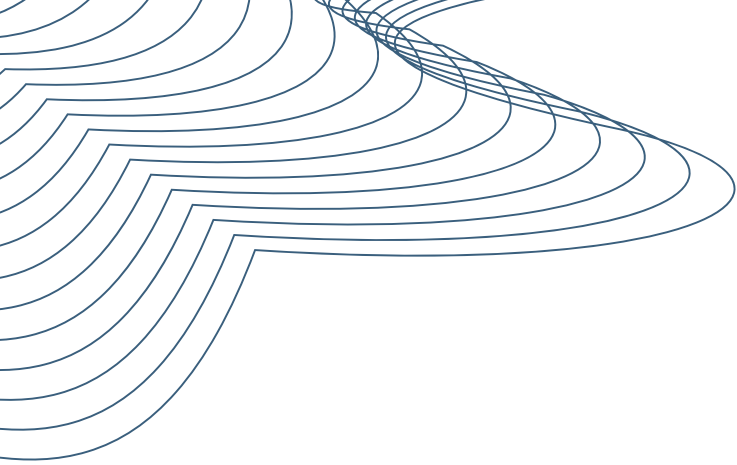
Specialization + Availability

Room

Type + Availability

Run Priority Scheduling + Resource Allocation

POSITION	PATIENT	PRIORITY	ARRIVAL	SPECIALIZATION	ROOM REQUIREMENT
#1	p117 - keerthi	4 - Critical	01:00	General Medicine	Operation Theatre
#2	p128 - meenakshi	4 - Critical	11:08	Cardiology	Operation Theatre
#3	p116 - ramnadham	4 - Critical	12:00	General Surgery	Operation Theatre



KL HOSPITAL

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9

Patients in Latest Run

2026-09-30 12:20:00

Run Time

Priority + Resources

Scheduling Method

POSITION	PATIENT	PRIORITY	DOCTOR	ROOM	TREATMENT	STATUS
#1	p117 keerthi	4 - Critical	Not available	Not available	Surgery	Waiting
#2	p128 meenakshi	4 - Critical	Not available	Not available	Surgery	Waiting
#3	p116 ramnadham	4 - Critical	Not available	Not available	Surgery	Waiting
#4	p113 prem sai	3 - High	D001 - Dr. Arun Kumar	Not required	Consultation	Scheduled
#5	p114 kichu	2 - Medium	D002 - Dr. Priya Sharma	Not required	Consultation	Scheduled
#6	p112 karthik	2 - Medium	D002 - Dr. Priya Sharma	Not required	Consultation	Scheduled
#7	p118 pawan	1 - Low	D006 - Dr. Vivek Das	MON-01 - Monitoring Room	Continuous Monitoring	Scheduled

Patients currently scheduled with their allocated resources.

Allocation rule: priority is the primary scheduling factor. Doctor matching uses specialization, availability and workload. Room matching uses the required room type and availability.

POSITION	PATIENT	PRIORITY	DOCTOR	ROOM	TREATMENT	ACTION
1	p244 sruthi	4 - Critical	D004 - Dr. Ananya Rao General Surgery	OT-02 Operation Theatre	Surgery	Discharge
1	p245 Ramkrishna	4 - Critical	D001 - Dr. Arun Kumar Cardiology	OT-01 Operation Theatre	Surgery	Discharge
2	p024 Thanu	3 - High	D006 - Dr. Vivek Das General Medicine	Not required	Consultation	Discharge
4	p113 prem sai	3 - High	D001 - Dr. Arun Kumar Cardiology	Not required	Consultation	Discharge
5	p114 kichu	2 - Medium	D002 - Dr. Priya Sharma Neurology	Not required	Consultation	Discharge
6	p112 karthik	2 - Medium	D002 - Dr. Priya Sharma Neurology	Not required	Consultation	Discharge
7	p118 pawan	1 - Low	D006 - Dr. Vivek Das General Medicine	MON-01 Monitoring Room	Continuous Monitoring	Discharge
8	p111 greeshma	1 - Low	D006 - Dr. Vivek Das General Medicine	Not required	Consultation	Discharge

CONCLUSION

The project provides a practical implementation of Priority Scheduling for hospital patient management. It determines the treatment order based on patient priority and manages doctor availability and room allocation. Through this project, we gain a better understanding of priority-based patient management, doctor coordination, and efficient use of hospital resources.



**THANK
YOU**

